

APEX INSTITUTE OF TECHNOLOGY & SCIENCES (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CONTINUOUS TEST 1 (CT-1) EXAMINATION • AY 2025-26

Course: Database Management Systems (CS501) | Faculty: Dr. Ravi Kumar

Date: 15 September 2025

Time: 60 Minutes

Max Marks: 20

Instructions: Answer all questions in Part A. Part B contains internal choice.

PART A — CORE CONCEPTUAL QUESTIONS

Answer all questions. Each question carries 2 marks.

1. Differentiate between Logical Data Independence and Physical Data Independence with a suitable example. [2 M]
2. Explain the three levels of DBMS architecture (ANSI/SPARC) and their significance. [2 M]
3. Define foreign key constraint and specify the effect of ON DELETE CASCADE. [2 M]

PART B — ANALYTICAL & DESIGN PROBLEMS

Answer both questions. Each carries 7 marks.

4. Design an Entity-Relationship (ER) diagram for a University Examination System showing cardinality ratios and primary keys. [7 M]
5. Convert the designed ER diagram into a set of Relational Tables, clearly identifying primary and foreign keys. [7 M]